Contrastive Tokens and Label Activation for Remote Sensing Weakly Supervised Semantic Segmentation

Zaiyi Hu, Junyu Gao, *Member, IEEE*, Yuan Yuan, *Senior Member, IEEE*, and Xuelong Li, *Fellow, IEEE*

***Abstract*— In recent years, there has been remarkable progress in weakly supervised semantic segmentation (WSSS), with vision transformer (ViT) architectures emerging as a natural fit for such tasks due to their inherent ability to leverage global attention for comprehensive object information perception. However, directly applying ViT to WSSS tasks can introduce challenges. The characteristics of ViT can lead to an oversmoothing problem, particularly in dense scenes of remote sensing images, significantly compromising the effectiveness of class activation maps (CAMs) and posing challenges for segmentation. Moreover, existing methods often adopt multistage strategies, adding complexity and reducing training efficiency. To overcome these challenges, a comprehensive framework Contrastive Token and Foreground Activation (CTFA) based on the ViT architecture for WSSS of remote sensing images is presented. Our proposed method includes a contrastive token learning module (CTLM), incorporating both patch-wise and class-wise token learning to enhance model performance. In patch-wise learning, we leverage the semantic diversity preserved in intermediate layers of ViT and derive a relation matrix from these layers and employ it to supervise the final output tokens, thereby improving the quality of CAM. In class-wise learning, we ensure the consistency of representation between global and local tokens, revealing more entire object regions. Additionally, by activating foreground features in the generated pseudo label using a dual-branch decoder, we further promote the improvement of CAM generation. Our approach demonstrates outstanding results across three well-established datasets, providing a more efficient and streamlined solution for WSSS. Code will be available at: https://github.com/ZaiyiHu/CTFA.**

***Index Terms*— Deep learning, remote sensing images, vision transformer (ViT), weakly supervised semantic segmentation (WSSS).**

I. Introduction

With the advancement of telemetry platform technologies, remote sensing image data holds a vast reservoir of valuable information that requires thorough exploration. Semantic segmentation, a pivotal task in computer vision, involves assigning specific categories to individual pixels within an image and finds extensive applications in areas like land cover mapping [3], intelligent agriculture [4], and natural environment detection [5]. However, the process of annotating pixels one by one to obtain segmentation labels is exceptionally time-consuming and labor-intensive, significantly hindering its practical utility. Consequently, there has been a growing focus on weakly supervised semantic segmentation (WSSS) techniques rooted in image-level supervision, which is the easiest label to gain. These approaches often entail utilizing a classification model to generate pseudo labels through class activation map (CAM) [6], subsequently training the segmentation model with these pseudo labels as ground truth [7], [8], which have achieved great performance so far.

Despite advancements in WSSS methods, challenges persist in the context of remote sensing imagery. On the one hand, remote sensing images exhibit distinct characteristics, including foreground-background imbalance, large-scale variations, and diverse multicategory scenes. These unique attributes pose considerable hurdles for precisely estimating CAM, making inaccurate segmentation results. As depicted in Fig. 1, compared with PASCAL VOC [9] and COCO [10], there is a noticeable decline in segmentation performance accompanies a reduction in the foreground proportion within the dataset. On the other hand, current weakly supervised segmentation

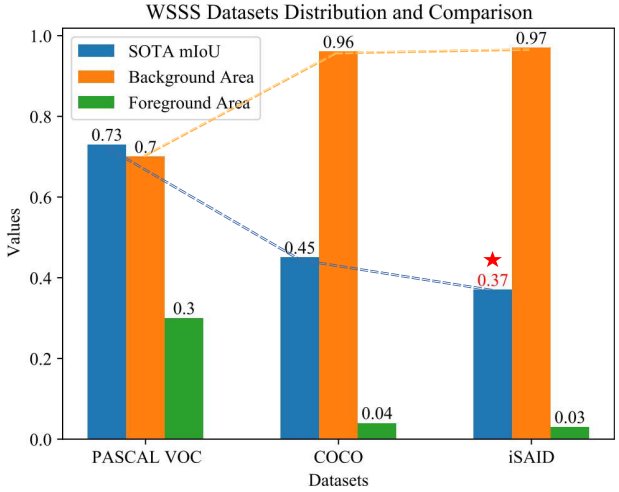


Fig. 1. Statistical comparison between WSSS datasets on distribution and performance. With lower foreground area, it becomes difficult for model to segment. The SOTA mIoU metrics in VOC and COCO come from [1] and [2], respectively.

Manuscript received 30 December 2023; revised 14 March 2024; accepted 23 March 2024. Date of publication 8 April 2024; date of current version 29 April 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0160400 and in part by the Natural Science Foundation of China under Grant 62306241 and Grant 61871470. *(Corresponding authors: Xuelong Li; Junyu Gao.)*

Zaiyi Hu and Yuan Yuan are with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: zaiyihu@gmail.com; y.yuan1.ieee@gmail.com).

Junyu Gao is with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China, and also with Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China (e-mail: gjy3035@gmail.com).

Xuelong Li is with the Institute of Artificial Intelligence (TeleAI), China Telecom Corporation Ltd., Beijing 100033, China (e-mail: xuelong_li@ieee.org).

Digital Object Identifier 10.1109/TGRS.2024.3385747

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

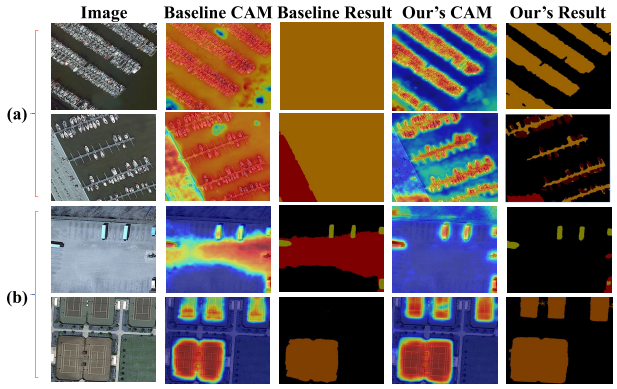


Fig. 2. Comparison between generated CAM and corresponding segmentation result before and after refinement. (a) Oversmoothing issue in ViT baseline. (b) Inaccurate foreground activation. The baseline here refers to vanilla ViT [13] architecture.

approaches in remote sensing often rely on complex multistage training strategies to improve the precision of pseudo label estimation [11], [12]. While effective, these strategies need to train multiple times, resulting in more intricate pipeline and higher training costs.

To address the outlined challenges, we propose an innovative solution that leverages a vision transformer (ViT)-based architecture [13] for single-stage WSSS, providing a more efficient and streamlined approach. Existing research has critiqued CAM for its inherent flaws, as it typically identifies only the most discriminative semantic regions, thereby compromising the overall performance of semantic segmentation [11], [14]. Furthermore, conventional convolutional neural networks (CNNs) are constrained by their inductive bias [15], [16], primarily capturing local features and struggling to reveal the entirety of an object. In contrast, ViT-based models, benefiting from a self-attention strategy, naturally establish global feature interactions, overcoming the limitations of CNN and uncovering entire object regions. It enhances global relation results in more accurate CAM generation, contributing to superior performance. However, directly applying ViT-based methods directly on WSSS for remote sensing data introduce two specific challenges.

A. Oversmoothing Issue

As depicted in Fig. 2(a), the CAM produced by the baseline ViT exhibits a tendency toward uniform activation over the entire area especially in a congested scene, indicating an oversmoothing issue. Unlike natural images, due to the large-scale differences and rich land information in remote sensing images, CAM is difficult to directly locate key objects and is more prone to overfitting. This behavior is mainly attributed to the low-pass filter characteristic of ViT [17], [18], which tends to reduce the variance of input signals. Consequently, image patch tokens become oversmoothed, resulting in indistinguishable CAM and adversely impacting the performance.

B. Inaccurate Foreground Activation

In Fig. 2(b), inaccuracies in foreground activation are evident. The crucial regions of objects are not accurately highlighted, with foreground regions being erroneously activated. This imprecision is detrimental to the effectiveness of WSSS.

To address these specific challenges, we introduce the contrastive token learning module (CTLM) and the label foreground activation module (LFAM). The main goal of CTLM is to highlight the key object regions and suppress the overactivated background regions, which comprises a patch-wise contrast module and a class-wise contrast module. Motivated by the observation that image patch representations in the earlier layers of ViT are typically more diverse [19], [20], the patch-wise contrast module supervises the final output patch tokens from ViT. It achieves this by incorporating token relations from intermediate layers, segregating different tokens based on their semantic labels into positive and negative pairs, and calculating their cosine similarity. By emphasizing larger positive pairs and smaller negative pairs, the representations of positive tokens become more consistent, while the negative tokens become more discriminative. This helps in creating more diverse representations, addressing the oversmoothing problem and producing more accurate CAM. The class-wise contrast module is employed to enhance semantic consistency between local nonobvious regions and global objects. This is achieved by minimizing the difference between class tokens embedding, facilitating the activation of more object regions in the CAM. Class tokens in ViT aggregate semantic information [13], [21], [22], crucial for the WSSS task to identify key object regions. To further refine the foreground activation, local images are randomly cropped in an auxiliary CAM to obtain the local images' class tokens. By maximizing the difference between class tokens of the global image and those of the local background image, discrepancies are introduced between the foreground and background, which helps in achieving more accurate foreground activation.

Moreover, given the notably diminutive size of foreground objects in aerial scenes, comprising minimal areas [23], the model encounters challenges in accurately perceiving key object regions, resulting in suboptimal CAM. Building on the concept presented in [24], which effectively activates features related to small objects, our proposed solution incorporates a LFAM to tackle this specific challenge. This module is implemented through a dual-branch decoder, comprising a segmentation branch and a foreground branch that produces semantic-level and binary-class outputs, respectively. Subsequently, we employ a cooperative loss mechanism to refine the generated pseudo label, enabling the model to discern foreground areas more effectively and thereby enhancing overall segmentation performance.

Our work innovatively integrates the aforementioned modules, leading to a successful enhancement in the quality of generated CAM and, consequently, improved segmentation results. Through the concurrent training of the ViT-based network with these modules within a single training cycle using image-level labels, our proposed WSSS method achieves heightened convenience and efficiency. The contributions of this article can be summarized as follows.

1) We propose a novel and effective ViT-based framework, contrastive token and foreground activation (CTFA) for the single-stage WSSS task on multiclass remote

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

sensing scenes. Experiments demonstrate our work achieves superb results on the previous benchmark for image-level WSSS tasks.

2) CTLM is proposed to address the issue of oversmoothing. This module incorporates both patch-wise contrastive learning and class-wise contrastive learning, thereby enhancing the overall quality of the CAM and improving segmentation performance.
3) LFAM is designed to activate the foreground area of the pseudo label, thereby enhancing the representation of key objects in the scene.

II. Related Work

A. Semantic Segmentation for Remote Sensing Scenes

Semantic segmentation for remote sensing scenes has become a pivotal task in image processing, garnering increasing attention with the rise of deep learning. Due to the diverse application scenarios and target categories, various subareas within remote sensing semantic segmentation employ similar techniques, including road extraction [25], [26], building extraction [27], [28], and saliency detection [5], [29].

CNNs and their variants have been widely adopted for remote sensing semantic segmentation tasks. For instance, Kuo et al. [30] employ a deep aggregation network based on fully convolutional networks [31] for land cover classification. Liu et al. [32] propose a model consisting of three CNNs for road extraction. With the evolution of network architectures, recent works in this area have integrated ViTs: He et al. [33] propose a mixed structure combining UNet and Swin Transformer, and Zhang et al. [34] utilizes a transformer with directional windows for semantic segmentation. Furthermore, not only different network design [35], advanced solutions are tailored to the characteristics of remote sensing images, such as multiscale representations [36] and attention mechanisms [37], have been employed to enhance segmentation performance.

Given that fully supervised learning has achieved a relatively high level, scholars are increasingly focusing on scenarios where high-quality annotated data are limited. This has led to research in weakly supervised [11], [38], [39], semisupervised [40], unsupervised [41], and few-shot learning [42]. These paradigms, to some extent, reduce the dependence on data and better meet the needs of real-world application scenarios.

B. WSSS for Remote Sensing Scenes

In contrast to pixel-level fully supervised segmentation tasks, weakly supervised labels provide less supervision information, often in the form of scribbles [43], points [44], bounding boxes [45], and image-level annotations [11]. Among these, image-level annotations have the lowest annotation cost. The use of image-level labels for semantic segmentation often relies on CAM to generate pseudo masks and supervise the learning of segmentation results. Many techniques incorporate additional cues extracted from the data, such as affinity information [7], [46] and co-occurrence matrix [12], [47].

In the field of remote sensing, WSSS has found widespread applications in various unique scenarios. Li et al. [48] adopt a multistage framework based on fully convolutional networks for building extraction. Fu et al. [49] propose WSF-Net, utilizing a feature fusion network and a balanced binary training strategy for cloud segmentation. As the WSSS process matures, the categories and scenarios become more diverse. Zhou et al. [11] employ a self-supervised Siamese network [8] and further propose a multicategory WSSS benchmark . They further propose a WSSS method via cross-image semantic mining, obtaining great results on the aforementioned benchmark [50]. Zeng et al. [51] propose a saliency map generator to produce location information, resulting in more accurate performance. OME utilize a series of preprocessing methods including co-occurrence and data filtering, achieving superior results approaching 90% of the performance of fully supervised counterparts [12].

While these methods significantly contribute to the development of WSSS for remote sensing scenes, current approaches in the remote sensing field often adopt a multistage paradigm for more accurate CAM generation and better performance. However, these models suffer from high complexity and require sophisticated training strategies [52]. To simplify the training process and better meet real-world demands, our method adopts a single-stage paradigm and utilizes contrastive learning to enhance the quality of CAM, thereby alleviating the challenges associated with single-stage training.

C. WSSS by ViTs

Instead of meticulously crafting network architectures, many WSSS approaches rely on common CNNs such as DeepLabv3+ [53], serving as the foundational feature extractor and CAM generator. These methods primarily concentrate on enhancing supervision, exemplified by works like 1stage [54], which employs pixel-adaptive mask refinement within the framework of DeepLabv3+ to facilitate robust learning from images. Similarly, SLRNet [52] leverages cross-view supervision and exploits low-rank properties to mitigate the impact of supervision errors.

As computer vision gains traction and backbone networks evolve, numerous concurrent endeavors integrate ViTs into the realm of WSSS, yielding notable performance gains. The inherent global awareness of ViT facilitates comprehensive object understanding. Xu et al. [21] introduced MCTformer, utilizing multiple class tokens to enhance localization maps. Building upon this, they introduced MCTformer+ [55], incorporating contrastive learning to augment the learning of class tokens. Ru et al. [46] proposed AFA, which exploits affinity information within the ViT framework. Extending their work, they introduced TOCO [19], addressing the oversmoothing issue by enforcing pairwise token similarity across various layers of the ViT.

Despite the considerable success of ViT in WSSS, limited efforts have been made to enhance it from the perspective of label activation in remote sensing. Our approach incorporates the LFAM, resulting in successful performance improvements.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

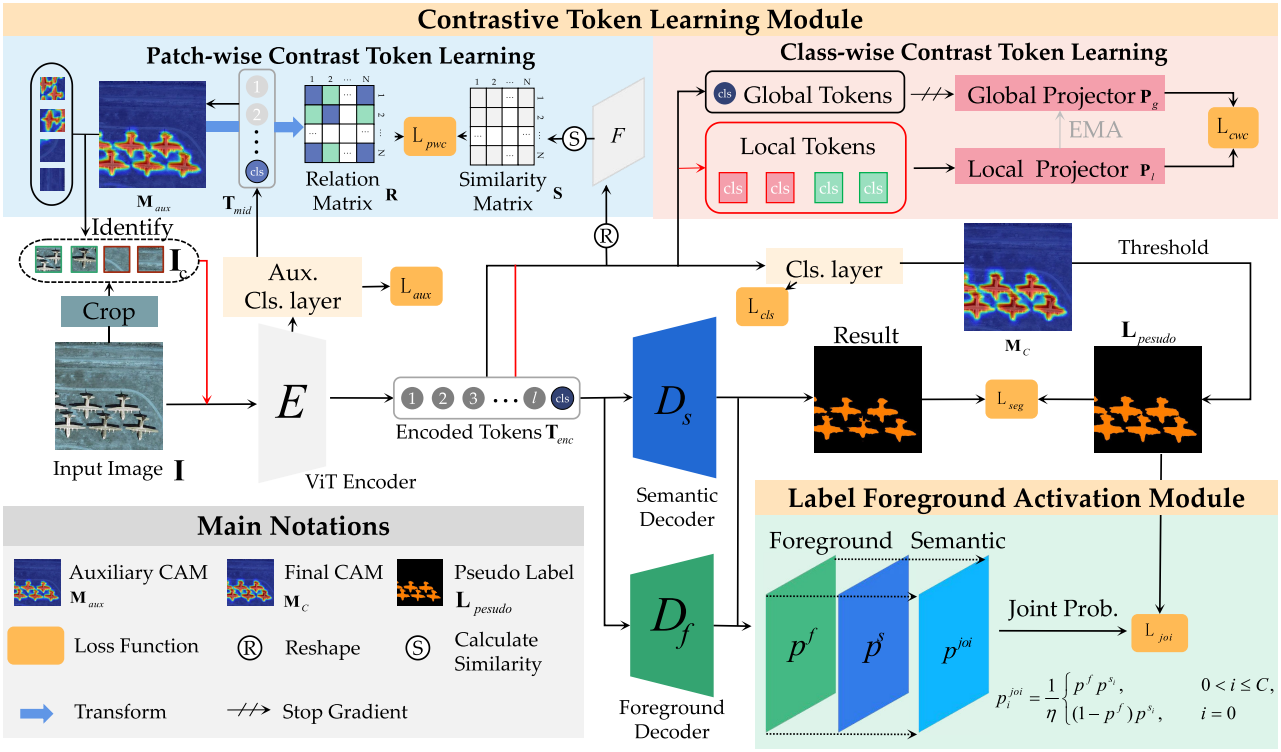


Fig. 3. Overview to our ViT-based WSSS framework, CTFA. Within the CTLM module, we incorporate patch-wise token contrast and class-wise token contrast. The former leverages CAM derived from the intermediate layer to generate a relation matrix, providing supervision for patch tokens in the final layer. Meanwhile, the latter exploits the consistency between class-token representations, contrasting class-token instances cropped from both local and global contexts. In the LFAM module, we calculate the joint loss of the dual-branch decoder, thereby facilitating the activation of foreground areas.

III. Method

A. Overview

As depicted in Fig. 3, the CTFA framework is built upon an encoder–decoder segmentation network leveraging the ViT architecture. The CTFA process initiates with an original input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, where H and W represent the height and width dimensions. Subsequently, the ViT encoder with patch size P divides the image $\mathbf{I}$ into patches and extracts features, resulting in a token sequence denoted as $\mathbf{T}_{\text{enc}} = [T^1_{\text{pat}}; T^2_{\text{pat}}; \ldots; T^l_{\text{pat}}; T_{\text{cls}}]$ with a length of $l = ((H \times W)/P^2)$. Following the conventions of previous WSSS frameworks [11], [12], we employ CAM [6] to derive the initial pseudo label $\mathbf{L}_{\text{pseudo}}$.

Specifically, the CAM is derived by summing the weights $\mathbf{W} \in \mathbb{R}^{c\times d}$ of a feature map $\mathbf{F} \in \mathbb{R}^{hw\times d}$ reconfigured into a 2-D shape from the output patches in a classification layer. For a specific class c, the CAM is defined as

$$\mathbf{M}_c = \sum_i \mathbf{W}_i^c \mathbf{F}_i, \quad \mathbf{CAM}_c = \frac{\text{Relu}(\mathbf{M}_c)}{\text{Max}(\text{Relu}(\mathbf{M}_c))} \tag{1}$$

where the Relu function is employed to eliminate negative activation. The pseudo label $\mathbf{L}_{\text{pseudo}}$ is then constructed based on **CAM** across all categories, utilizing two background thresholds τ_l and τ_h $(0 < \tau_l < \tau_h < 1)$ to distinguish between foreground, background, and uncertain regions. For each element at position x, y, using CAM on total categories $\mathbf{M}_C^{x,y}$ for calculation as follows:

$$\mathbf{L}^{x,y}_{\text{pseudo}} = \begin{cases} 0, & \text{if } \max\left(\mathbf{M}_C^{x,y}\right) \le \tau_l \\ \text{argmax}\left(\mathbf{M}_C^{x,y}\right), & \text{if } \max\left(\mathbf{M}_C^{x,y}\right) \ge \tau_h \\ 255, & \text{otherwise} \end{cases} \tag{2}$$

where and 0 and 255 represent the indices of the background and uncertain regions and C represent the total categories in the dataset, respectively. Pixel-adaptive refinement (PAR) module [46] is also adopted on $\mathbf{L}_{\text{pseudo}}$ for further enhancing. This pseudo label is then utilized as a supervision for the segmentation probability outcome $p^s \in \mathbb{R}^{H\times W}$ using cross-entropy to calculate the segmentation loss $\mathcal{L}_{\text{seg}}$.

In addition to the main pipeline described above, the framework is further enhanced by the inclusion of the CTLM and the LFAM, which will be elaborated in detail in Sections III-B and III-C.

B. Contrastive Token Learning Module

1) Patch-Wise Token Contrastive Learning: Leveraging the diversity among patch representations in different layers of ViT, the primary aim of Patch-Wise Token Contrastive Learning is to induce distinct semantic characteristics, thus alleviating the oversmoothing problem in ViT [19], [20].

This is achieved by introducing a contrastive loss between the patch similarity $\mathbf{S}(\mathbf{T}_{\text{enc}})$ and the relation matrix $\mathbf{R}(\mathbf{T}_{\text{mid}})$ extracted from the intermediate layer of ViT. Observing that intermediate layers of ViT still retain some semantic diversity, the cosine similarity in the early layers of the ViT block is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

expected to be lower than in the later ones, denoted as follows:

$$\mathbf{S}(\mathbf{T}) = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{\left|\mathbf{T}_i{}^{\mathsf{T}} \mathbf{T}_j\right|}{\|\mathbf{T}_i\|_2 \left\|\mathbf{T}_j\right\|_2} \tag{3}$$

where the patch tokens $\mathbf{T} = [T'^1_{\text{pat}}; T'^2_{\text{pat}}; \ldots; T'^l_{\text{cls}}]$ are sourced from different layers of ViT. After obtaining the patch token sequence $\mathbf{T}_{\text{enc}}$ from the final layer of the ViT encoder, we reshape it into a 2-D feature map $\mathbf{F}$. Subsequently, absolute cosine similarity matrix $\mathbf{S}(\mathbf{T}_{\text{enc}})$ is constructed according to (3).

Meanwhile, an auxiliary CAM $\mathbf{M}^{\text{aux}}$ is then produced based on the intermediate tokens $\mathbf{T}_{\text{mid}}$ derived from the intermediate layer, which is set as the ninth layer following conventional configurations [19]. Using (2), the generated CAM $\mathbf{M}_{\text{aux}}$ is then transformed into label $\mathbf{L}_{\text{mid}} \in \mathbb{R}^{\times h' \times w'}$ and flatten into a sequence $\mathbf{S}_{\text{mid}} \in \mathbb{R}^{1 \times h' w'}$. By repeating $h'w'$ times and stacking them together, a label matrix is constructed. Transposing this matrix and comparing it with the original yields the relation matrix $\mathbf{R}(\mathbf{T}_{\text{mid}})$, where each element is labeled positive for shared category information and negative otherwise.

Positive masks Mask_{pos} and negative masks Mask_{neg} of $\mathbf{R}(\mathbf{T}_{\text{mid}})$ are then extracted to maximize the similarity for positive pairs and minimize for negative ones. By summing the Hadamard product $\odot$ of masks and similarity, the loss function is formulated as

$$\mathcal{L}_{\text{pwc}} = \frac{1}{2}\left(1 - \frac{\sum_{i,j} \text{Mask}_{\text{pos}} \odot \mathbf{S}(\mathbf{T}_{\text{enc}})_{i,j}}{N^+}\right) + \frac{1}{2}\left(\frac{\sum_{i,j} \text{Mask}_{\text{neg}} \odot \mathbf{S}(\mathbf{T}_{\text{enc}})_{i,j}}{N^-}\right) \tag{4}$$

where N^+ and N^- represent the number of pairs Mask_{pos} and Mask_{neg}, respectively.

2) Class-Wise Token Contrastive Learning: The class-token is a crucial component in ViT, as its representation consolidates semantic information from key objects [19], [21], [55]. By ensuring the semantic consistency between global and local class-token representations, we unveil more semantic regions, leading to the activation of more complete objects.

In practice, we start by utilizing the auxiliary CAM $\mathbf{M}^{\text{aux}}$ and converting it into a reliable label using (3). After identifying uncertain and background regions in the label, we intentionally crop these areas to gather local images $\mathbf{I}^c \in \mathbb{R}^{H^c \times W^c \times 3}$ from the original input image $\mathbf{I}$. These cropped images are then input into the ViT encoder, producing smaller dimensional tokens that incorporate additional classification information. To ensure consistency between local and global representations, two projection heads $\mathbf{P}_l$ and $\mathbf{P}_g$ consisting of linear layers receive the two sets of tokens. The gradient of the global projection head $\mathbf{P}_g$ is halted, and an exponential moving average (EMA) is employed for its update. This helps smoothing the parameter updates of the projection heads, promoting stable model training.

Consider the projected global class token $\mathbf{T}^{\mathbf{P}}_{\mathbf{g}}$ and projected local class tokens $\mathbf{T}^{\mathbf{P}}_{\mathbf{l}^+}$, $\mathbf{T}^{\mathbf{P}}_{\mathbf{l}^-}$ from the uncertain and background regions, the loss function is formulated using InfoNCE loss [56] as follows:

$$\mathcal{L}_{\text{cwc}} = \frac{1}{N_{l^+}} \sum_{t_{l^+}} \log \frac{e^{\left(t_g^{\mathsf{T}} t_{l^+}/\tau\right)}}{e^{\left(t_g^{\mathsf{T}} t_{l^+}/\tau\right)} + \sum_{t_{l^-}} e^{\left(t_g^{\mathsf{T}} t_{l^-}/\tau\right)}} \tag{5}$$

where N_{l^+} means the number of $\mathbf{T}^{\mathbf{P}}_{\mathbf{l}^+}$, and $t_{l^+} \in \mathbf{T}^{\mathbf{P}}_{\mathbf{l}^+}$, $t_{l^-} \in \mathbf{T}^{\mathbf{P}}_{\mathbf{l}^-}$, $t_{g^+} \in \mathbf{T}^{\mathbf{P}}_{\mathbf{g}}$, respectively. Through the contrastive learning in (5), the class-token representations of the global object and local uncertain regions become more closely related, while the background regions are suppressed. This encourages the activation of more object regions.

Overall, the final object loss of CTLM is calculated as

$$\mathcal{L}_{\text{CTL}} = \lambda_1 \mathcal{L}_{\text{pwc}} + \lambda_2 \mathcal{L}_{\text{cwc}} \tag{6}$$

where λ_1 and λ_2 are two balance factors.

C. Label Foreground Activation Module

Inspired by the concept of a dual-branch decoder for both foreground and semantic predictions [24], we optimize the joint loss with the semantic output $p^s \in \mathbb{R}^{H \times W}$ and foreground output $p^f \in \mathbb{R}^{H \times W}$ to activate class-agnostic foreground objects, promoting the revelation of entire objects. As no specific labels are provided in the WSSS task, we naturally focus on optimizing the created pseudo label $\mathbf{L}_{\text{pseudo}}$.

In our implementation, we introduce an additional decoder $\mathbf{D}_f$ for the extracted features $\mathbf{T}_{\text{enc}}$ to generate the foreground output p^f. Considering the output probability p^f, the background probability is then denoted as $1 - p^f$ since it predicts only these two categories. Similarly, assuming output probability p^s from the semantic decoder $\mathbf{D}_s$, the background and foreground representations are p^{s_0} and $p^{s_i > 0}$, $(0 < i \leq C)$. The predictions from both branches are then combined to form the joint probability p_i^{joi}

$$p_i^{\text{joi}} = \frac{1}{\eta} \begin{cases} p^f p^{s_i}, & 0 < i \leq C \\ (1 - p^f) p^{s_i}, & i = 0 \end{cases} \tag{7}$$

where η is a normalizing factor, representing the summation of unmatched prediction outputs: $\eta = p^f p^{s0} + (1 - p^f) \sum_{i=1}^{C-1} p^{s_i}$. Finally, assuming class-wise ground truth $y_i \in 0, 1$ from the pseudo label $\mathbf{L}_{\text{pseudo}}$, where p_i^{joi} is for class i with label $y_i = 1$, the joint loss is given by an expansion of cross-entropy loss

$$\mathcal{L}_{\text{joi}} = -\sum_{i=0}^{C-1} y_i \log\left(p_i^{\text{joi}}\right). \tag{8}$$

D. Loss Functions

In addition to the loss in CTLM $\mathcal{L}_{\text{CTL}}$, the loss in LFAM $\mathcal{L}_{\text{joi}}$, and the segmentation loss $\mathcal{L}_{\text{seg}}$, CTFA also includes classification losses $\mathcal{L}_{\text{aux}}$ and $\mathcal{L}_{\text{cls}}$, as illustrated in Fig. 3. Both classification losses are based on the multilabel soft margin loss. The overall total loss function of CTFA is the sum of the aforementioned terms,

$$\mathcal{L}_{\text{CTFA}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{aux}} + \mathcal{L}_{\text{CTL}} + \lambda_3 \mathcal{L}_{\text{joi}} + \mathcal{L}_{\text{seg}} \tag{9}$$

where λ_3 is a balancing factor for LFAM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON WITH WSSS METHODS ON THE iSAID DATASETS. THE METHODS WITH * INDICATES MULTISTAGE WSSS METHODS

Model	BG	GTF	SBF	SV	SH	BR	BC	BD	RA	PL	TC	LV	ST	HA	SP	HC	mIoU
AffinityNet [CVPR, 2017] * [7]	77.75	14.70	43.72	3.69	5.77	6.32	18.81	20.40	9.76	7.53	54.25	23.78	42.43	3.95	1.84	1.00	20.82
IRNet [CVPR,2019] * [61]	76.62	18.88	39.82	6.72	16.62	7.48	24.47	16.35	5.69	22.02	54.91	28.96	29.60	9.81	1.31	3.01	22.64
CIAN [AAAI, 2020] * [62]	78.30	21.90	40.80	8.90	17.50	8.30	28.40	22.70	13.50	18.10	47.50	27.60	30.20	8.50	4.20	5.30	23.90
SEAM [CVPR, 2020] * [8]	73.20	28.40	37.80	4.70	15.30	6.90	43.10	31.40	16.70	19.60	44.30	31.10	24.40	7.80	9.80	8.30	24.50
SC-CAM [CVPR, 2020] * [63]	76.23	28.55	36.07	11.11	19.95	5.67	23.07	36.98	16.01	24.78	54.32	30.07	29.89	10.29	17.34	11.92	26.62
RPNet [TMM, 2021] * [64]	80.20	33.60	42.10	9.80	16.90	7.20	45.20	32.60	26.10	27.60	51.50	29.40	42.70	9.30	16.80	13.70	30.30
blueS^{2}EPC [TGRS, 2022] * [11]	86.11	38.30	46.70	11.48	17.10	8.71	**53.30**	36.68	**34.59**	31.70	56.74	30.20	**49.10**	11.59	19.00	15.86	36.79
blueCISM [RS, 2023] * [50]	83.10	33.50	44.60	11.80	19.50	8.40	48.30	34.40	29.20	30.50	52.60	31.90	44.10	10.80	17.60	14.30	32.20
blueOME [TGRS, 2023] * [12]	**94.62**	36.60	**52.72**	13.12	22.53	13.12	32.49	9.87	31.15	**48.38**	**66.22**	**54.73**	43.03	31.31	8.17	16.99	35.94
1stage [CVPR, 2020] [54]	71.37	31.38	26.51	2.09	4.19	12.64	52.23	17.91	13.80	22.66	43.54	6.87	36.58	4.99	15.88	0.00	22.66
SLRNet [IJCV,2022] [52]	74.66	38.13	30.69	2.18	14.10	7.36	33.92	24.40	23.95	16.46	44.11	39.23	30.21	8.20	12.56	12.46	25.79
AFA [CVPR, 2022] [46]	75.17	31.50	40.13	4.76	8.68	2.12	41.26	**48.86**	13.79	24.29	47.25	39.53	43.68	14.22	29.02	**25.03**	30.58
TOCO [CVPR, 2023] [19]	90.17	54.60	39.52	11.85	27.98	14.93	29.55	38.94	13.56	37.44	48.11	33.20	39.46	32.46	32.22	23.24	35.45
CTFA [Ours]	90.20	**55.97**	40.18	**14.87**	**28.73**	**14.97**	42.12	38.21	20.15	37.72	50.67	35.72	48.15	**32.85**	**32.82**	22.10	**37.84**

It is noteworthy that since the ground truth in LFAM is simultaneously generated and updated with the entire network, both the joint loss $\mathcal{L}_{\text{joi}}$ and segmentation loss $\mathcal{L}_{\text{seg}}$ are retained for optimization. By integrating these modules into a single-stage WSSS framework, our method also incorporates a regularization loss term for spatial consistency of segmentation masks, following previous works [52].

IV. EXPERIMENTS & RESULTS

A. Descriptions of Datasets

To comprehensively assess the effectiveness of our method, we conduct experiments on three commonly employed remote sensing datasets. The detailed information about these datasets and our implementation strategies are described as follows.

1) iSAID Dataset: The iSAID dataset [23], designed for both instance segmentation and semantic segmentation in aerial images, is a substantial collection adapted from the object detection dataset DOTA [57]. Comprising 2806 optical remote sensing images, it encompasses 16 categories, including background, ground track field, soccer ball field, small vehicle, ship, bridge, basketball court, baseball diamond, roundabout, plane, tennis court, large vehicle, storage tank, harbor, swimming pool, and helicopter. The category names listed above correspond to the abbreviations in Table I. Following the data partitioning methodology outlined in [11], we perform random cropping of 512 × 512 tiles from each image. This process results in a training set comprising 7500 tiles, a validation set with 1653 tiles, and a test set with 1315 tiles.

2) Potsdam Dataset: The ISPRS Potsdam [58] is an aerial image dataset captured in the city of Potsdam. It comprises 38 orthorectified images, each with dimensions of 6000 × 6000 pixels and a spatial resolution of 5 cm. The dataset encompasses six categories, namely impervious surfaces, buildings, low vegetation, trees, cars, and clutter. We use 23 images allocated for training, and 14 images reserved for testing. To create manageable tiles, we crop these images into nonoverlapping patches of 512 × 512 pixels. This process results in a training set of 2783 tiles and a testing set of 1794 tiles.

3) DeepGlobe Land Cover Dataset: The DeepGlobe dataset [59] comprises 830 images, each with an average size of 2448 × 2448 pixels. The dataset includes seven categories: urban, agriculture, rangeland, forest, water, barren, and unknown. Similar to previous datasets, we crop these images into nonoverlapping patches of size 512 × 512 pixels. This process results in a training set with 9100 slices, a testing set with 2025 slices, and a validation set with 2150 slices.

B. Implementation Details

We opted to utilize the vanilla Vision Transformer (ViT-B) [13] as our baseline. During the training stage, we employed an AdamW optimizer on a single NVIDIA RTX 3090 GPU for 20 000 iterations with a constant batch size of 8. The weight decay is set to 1e-2, and the momentum is set to 0.9. The initial learning rate is established at 6e-5, except for the DeepGlobe dataset, where a learning rate of 2e-5 is used. Learning rate decay followed a polynomial scheduler. For the decoder part, we implemented a simple segmentation head as described in [19], consisting of two dilated convolution layers with a dilation rate of 5. During the inference stage, a multiscale test strategy is adopted to generate multiscale CAM, which are fused by averaging to obtain the final pseudo label. Subsequently, dense CRF [60] is employed as a postprocessing step to refine the final result. Further details regarding the impact of key hyperparameters will be discussed in Section IV-E.

C. Evaluation Metrics

Following the previous evaluation process for semantic segmentation [11], [12], we primarily utilize the mean intersection over union (mIoU) to assess performance. Additionally, $F1$ scores and overall accuracy (OA) are employed for a more comprehensive evaluation of our method. These metrics are typically defined as follows:

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \tag{10}$$

$$F1 = \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}} \tag{11}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FN}+\text{FP}+\text{TN}}. \tag{12}$$

Here, TP, FP, TN, and FN represent the count of true positives, false positives, true negatives, and false negatives, respectively. The metrics Precision and Recall are defined as $\text{TP}/(\text{TP}+\text{FP})$ and $\text{TP}/(\text{TP}+\text{FN})$. These metrics collectively provide a robust evaluation of the segmentation performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

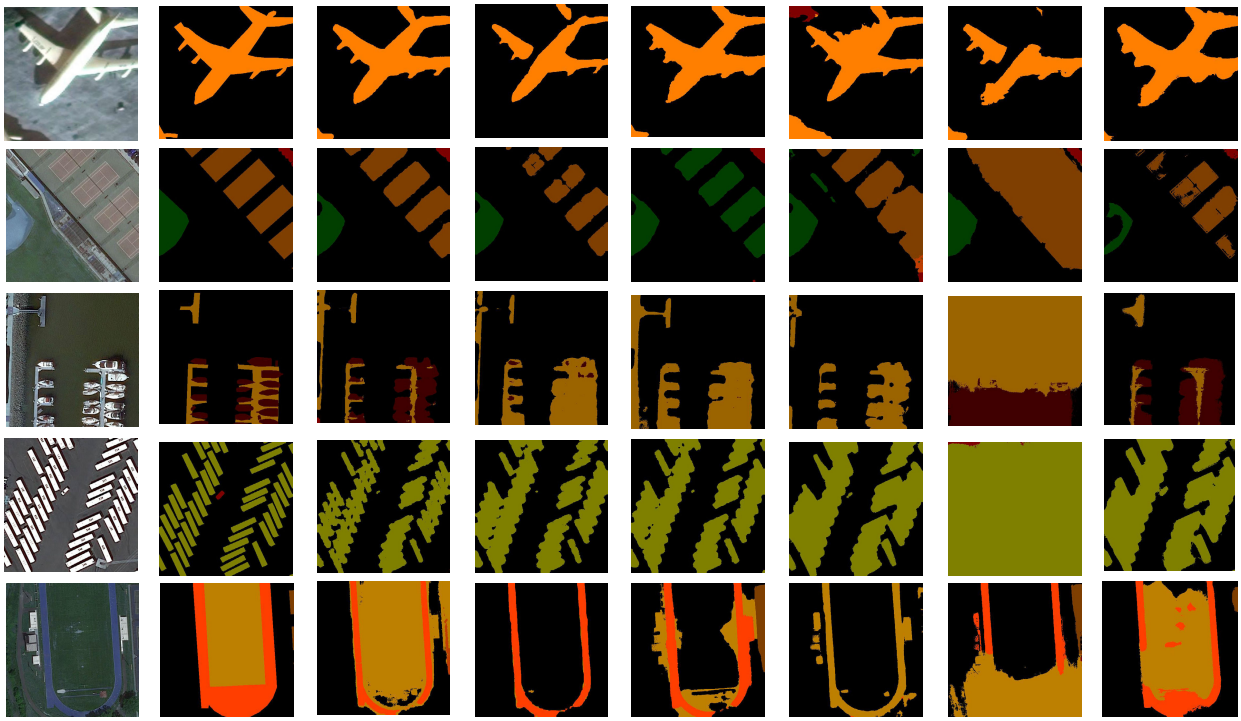


Fig. 4. Visualization results of the WSSS methods on the iSAID datasets, including categories as PL,TC, BD, SV, HA, SH, LV, BC, and GTF. (a) Input image. (b) Ground truth. (c) Ours. (d) TOCO. (e) AFA. (f) SLRNet. (g) 1stage. (h) OME.

D. Comparison With SOTA Methods

In this section, we present the results of state-of-the-art (SOTA) methods to facilitate a comprehensive performance comparison. Following the benchmark set by [11], four additional methods are included in the evaluation: 1stage [54], SLRNet [52], AFA [46], and TOCO [19]. The learning rates and other hyperparameters are adjusted based on their performance across the three remote sensing datasets.

1) iSAID Dataset: The iSAID dataset is recognized for its challenges and diverse categories, including small and dense objects like SV and SH. As shown in Table I, our method demonstrates notable success in handling small, dense objects, outperforming the second-best method TOCO [19] by 3.02% mIoU for SV and 0.75% mIoU for SH, respectively. Notably, the results in Table I imply that multistage methods generally outperform single-stage methods. In addition, we also reproduce the results of OME method under the same experimental setting [12] on the iSAID dataset, which has achieved remarkable metrics on the PL, TC, and LV categories. However, our single-stage method achieves the highest mIoU compared to multistage methods, validating the effectiveness of our design.

Regarding the visualization results, as depicted in Fig. 4, it is evident that the segmentation results by our method are more accurate, with the entire region activated. The results effectively showcase the efficiency of our model in two aspects.

1) In the segmentation results of PL and GTF, more complete regions have been discovered with explicit boundaries.
2) In the segmentation result of dense objects like vehicles and SH, more accurate foreground regions have been activated with fewer overlapping pixels. This demonstrates that our method achieves excellent results due to the benefits of the proposed modules.

2) Potsdam Dataset: The accuracy comparison with WSSS methods on the Potsdam dataset is presented in Table II. The results indicate that our method achieves superior performance on all metrics across almost every category. The overall mIoU improves by 4.64% compared to TOCO [19], with the highest mIoU attained in the clutter and car category. This is particularly noteworthy, as they are challenging categories with a low proportion in the dataset. The visualization results in Fig. 5 further illustrate the efficiency of our design. We successfully identify the clutter circle in the first row compared to other methods. Additionally, the segmentation result from our method exhibits robustness when facing the dim side of the building in the fourth row, while other methods may encounter issues with not activating the entire object region.

3) DeepGlobe Land Cover Dataset: Compared with the aforementioned datasets iSAID and Potsdam, the DeepGlobe dataset is unique as it lacks specific foreground and background differences [65]. It is primarily designed for large-area classification rather than precise focusing on small objects. This underscores the importance of our model having a better awareness of general object regions. The comparison results are presented in Table III, revealing that our segmentation results outperform other WSSS methods on all metrics. As for the visualization results in Fig. 6, it is evident that our model has explicitly identified the edge of the rangeland area, while other methods may face difficulties in accurately activating the rangeland region. Despite the differences from traditional WSSS datasets, our method still achieves a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON WITH WSSS METHODS ON THE POTSDAM DATASETS

Model	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	mIoU	OA	F1 Score
1stage [CVPR, 2020] [54]	37.41	42.05	34.25	23.17	21.50	8.50	27.81	43.52	40.16
SLRNet [IJCV, 2022] [52]	51.20	45.93	35.86	33.61	39.63	4.63	35.14	54.12	48.12
AFA [CVPR, 2022] [46]	58.60	48.94	45.93	27.40	42.13	16.68	39.95	60.15	51.21
TOCO [CVPR, 2023] [19]	53.45	**50.89**	47.72	45.21	48.21	6.34	41.97	64.25	54.21
CTFA [Ours]	**58.75**	50.58	**51.20**	**50.35**	**50.82**	**17.96**	**46.61**	**66.23**	**57.12**

TABLE III
COMPARISON WITH WSSS METHODS ON THE DEEPGLOBE DATASETS

Model	Urban	Agriculture	Rangeland	Forest	Water	Barren	Unknown	mIoU	OA	F1 Score
1stage [CVPR, 2020] [54]	47.87	50.90	14.76	58.62	38.73	29.43	21.54	37.41	60.12	51.80
SLNet [IJCV, 2022] [52]	64.07	70.55	10.04	52.01	**68.35**	45.80	15.48	46.61	73.45	62.72
AFA [CVPR, 2022] [46]	67.40	56.90	23.53	77.62	29.19	50.81	33.15	48.37	72.42	61.18
TOCO [CVPR, 2023] [19]	69.28	**80.57**	25.08	67.56	55.87	46.06	24.15	52.65	74.25	62.52
CTFA [Ours]	**70.02**	80.11	**30.07**	**69.90**	56.98	**50.82**	**35.14**	**56.15**	**76.12**	**64.12**

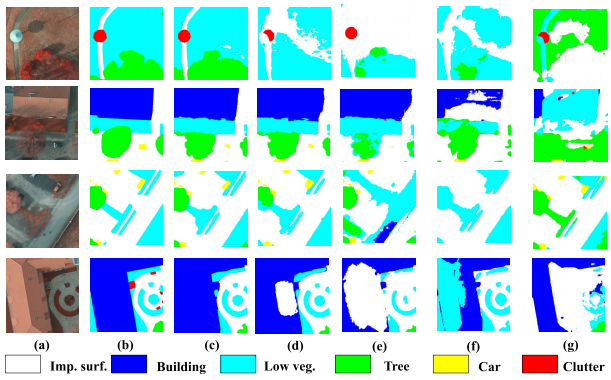


Fig. 5. Visualization results of the WSSS methods on the Potsdam datasets. (a) Input image. (b) Ground truth. (c) Ours. (d) TOCO. (e) AFA. (f) SLRNet. (g) 1stage.

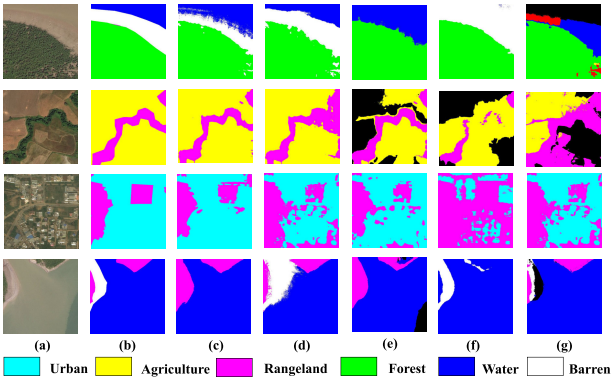


Fig. 6. Visualization results of the WSSS methods on the deepGlobe datasets. (a) Input image. (b) Ground truth. (c) Ours. (d) TOCO. (e) AFA. (f) SLRNet. (g) 1stage.

relatively acceptable result, emphasizing its adaptability and effectiveness across diverse scenarios.

E. Ablation Studies

We mainly discuss the effectiveness of our proposed modules, conducting ablation study to explore how our design affects the pseudo label, CAM and segmentation result.

1) Effects of the Patch-Wise Contrast: The primary objective of Patch-Wise Contrast is to leverage the intermediate layers of the transformer, which are known to exhibit lower similarity across patches [17], [18]. This intermediate layer is used to supervise the already highly similar final layer's output patches. This approach alleviates the patch-wise cosine similarity, resulting in improved CAMs and better segmentation performance. To further elucidate this issue, we calculate the average absolute cosine similarity across all 12 transformer blocks using the (3). This involves obtaining the output of each transformer block, calculating pair-wise cosine similarity for each token, and then determining the mean value. The statistical results in Table IV and the CAM visualizations in Fig. 7 confirm the efficiency of the patch-wise contrast modules. As suggested in Table IV, with PWC, the baseline result with auxiliary CAM has improved sharply by 10.09% mIoU, and the overall mIoU performance has increased by 11.89% mIoU, signifying a substantial improvement. Simultaneously, in Fig. 7, the Baseline with PWC has noticeably reduced the oversmoothing issue occured in congested sences.

Moreover, as depicted in Fig. 8, the results show that the original ViT-B exhibits a steadily increasing trend across all stages, with similarity values escalating as the transformer goes deeper, ultimately reaching 0.91 in the last block. In contrast, our method with Pixel-Wise Contrast effectively mitigates the overall similarity, resulting in a decrease of 0.2 in the cosine similarity.

To summarize, three substantial evidences show that PWC effectively address the oversmoothing issue.

1) A substantial enhancement in CAM and segmentation performance is observed in ablation experiments.
2) Visualization of the ablation experiment demonstrates a notable reduction in oversmoothing when PWC is incorporated.
3) Analysis of cosine similarity across transformer blocks shows a significant decrease with PWC, validating its effectiveness in mitigating oversmoothing. All these phenomena collectively validate the success of the PWC module.

2) Effects of the Class-Wise Contrast: Class token is initially designed for classification tasks in the original ViT [13], and it has been observed by researchers to carry semantic information about images [21]. Through the contrastive learning between class tokens from the global image and local crops, the model can reveal more accurate foreground regions. The results from Table IV illustrate that CWC significantly enhances the quality of CAMs, resulting in an overall improve-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

ABLATION STUDY OF THE PROPOSED MODULES ON THE iSAID DATASETS. THE BASELINE HERE REFERS TO THE VIT-B [13]

Method	CTLM		LFAM	CAM	Aux CAM	Small Objects mIoU		mIoU
	PWC	CWC				Small vehicle	Ship	
Baseline	-	-	-	26.54	-	8.12	15.17	22.45
Baseline + Aux CAM	-	-	-	22.72	23.42	7.91	15.23	23.62
Baseline + Aux CAM	√	-	-	32.81	31.70	12.01	26.84	35.51
	√	√	-	36.47	33.41	12.18	27.16	36.79
	√	√	√	**38.95**	**33.61**	**14.87**	**28.73**	**37.84**

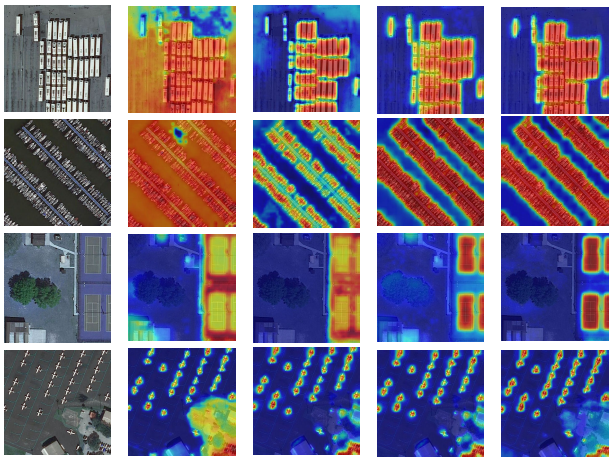


Fig. 7. Visualization results of the generated CAM on the iSAID datasets. (a) Input image. (b) Baseline. (c) Baseline + PWC. (d) Baseline + CTLM. (e) Baseline + CTLM + LFAM (ours).

ment in segmentation performance. The CAM visualizations in Fig. 7 further support our design. In the (d) column, the CAM becomes more focused on the entire activated region, showing high attention toward the target objects like ships and vehicles.

3) Effects of the Label Foreground Activation: Through the dual-branch decoder designed to decode the foreground and semantic information of the pseudo label, detailed regions of small objects have been more effectively revealed. As shown in Table IV, the best performance has been achieved with the proposed LFAM. Furthermore, the segmentation performance of small objects, such as small vehicles and ships, has improved by 2.69% and 1.57%, respectively. The CAM results in Fig. 7 also highlight the enhanced awareness of small objects in remote sensing images, such as ships or planes.

Additionally, an additional experiment in Table V on the Potsdam dataset evaluates the performance of generated pseudo labels and makes a comparison between the proposed modules. Compared with our final module, the initial pseudo label performance has decreased by 13.88%, showcasing significant improvement from our design.

In conclusion, experimental observations highlight the LFAM's ability for enhancing foreground activation: 1) segmentation performance particularly for small objects significantly improves with LFAM and 2) visualization in Fig. 7 demonstrates LFAM's superior performance in capturing detailed regions of small objects such as ships or planes.

4) Hyperparameter Setting: Some crucial hyperparameter setting are discussed in this subsection. The impact of the balance factor λ_3 for LFAM in Section III-C is discussed in

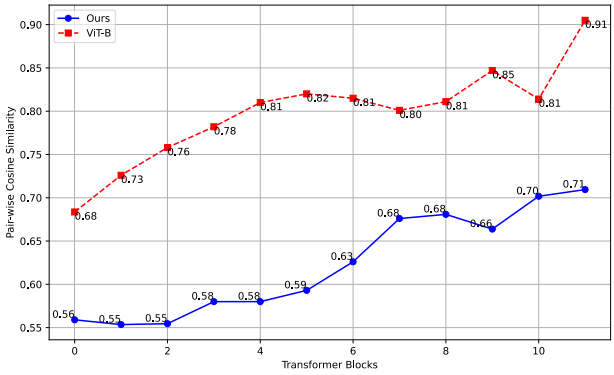


Fig. 8. Average pair-wise absolute cosine similarity of 12 transformer blocks obtained on the iSAID dataset.

TABLE V

ABLATION STUDY OF THE PROPOSED MODULES AND PESUDO LABEL QUALITY ON THE POTSDAM DATASET

Datasets	CTLM	LFAM	Label mIoU	mIoU	OA	F1
Potsdam	-	-	33.41	32.64	64.12	52.78
	√	-	39.42	40.84	70.56	58.16
	-	√	36.43	35.15	72.45	62.48
	√	√	**47.29**	**46.61**	**76.12**	**64.12**

TABLE VI

IMPACT OF THE BALANCE FACTOR λ_3 FOR LFAM

λ_3	0.05	0.1	0.2	0.3
mIoU	36.82	37.12	**37.84**	37.04

TABLE VII

IMPACT OF THE GLOBAL CROP SIZE AND LOCAL CROP SIZE FOR THE CLASS-WISE CONTRAST MODULE

Local Crop \ Global Crop	224^2	320^2	448^2
80^2	30.51	32.73	34.84
96^2	31.84	**37.84**	37.40
112^2	33.79	36.76	37.75

TABLE VIII

COMPUTING RESOURCES EVALUATION OF DIFFERENT NETWORKS. TIME REFERS TO THE TIME COST OF A CYCLE IN MODEL'S FORWARD PROPAGATION

Method	Backbone	Params (M)	GFLOPs	Time (ms)	mIoU
1stage [CVPR, 2020] [54]	WideResnet38	138	67.91	76.12	22.66
SLRNet [IJCV, 2022] [52]	WideResnet38	173	135.91	**73.75**	25.79
AFA [CVPR, 2022] [46]	MiT-B	**14**	**19.36**	121.39	30.58
TOCO [CVPR, 2023] [19]	ViT-B	92	93.79	120.39	35.45
CTFA + Pyramid Decoder	ViT-B	125	128.02	176.84	**37.91**
CTFA + ASPP Decoder	ViT-B	103	110.62	150.78	34.51
CTFA (Ours)	ViT-B	**97**	**99.84**	**138.48**	37.84

Table VI. Since previous similar balance factors λ_1, λ_2 are dedicatedly explored and set as 0.2 and 0.5 in [19], it's crucial to further validate the value of λ_3. Similarly, we also follow their setting to set the background thresholds τ_l and τ_h as 0.25 and 0.7. As shown in Table VI, we show that $\lambda_3 = 0.2$ obtain the highest performance. The global crop size and local crop size is listed in Table VII. Remote sensing images have

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
COMPARISON BETWEEN FULLY SUPERVISED COUNTERPARTS OF THE WSSS METHODS ON iSAID DATASETS

Method	Backbone	F	I	Ratio
1Stage	WideResnet38	42.42	22.66	53.42%
SLRNet	WideResnet38	42.45	25.79	60.75%
AFA	MiT-B	45.26	35.45	**78.32%**
CTFA	ViT-B	**48.57**	**37.84**	77.90%

many small foreground regions and an adequate crops should fully include the key objects while retain less unrelated object regions. Therefore, the local crops and global crops are set as 96^2 and 320^2 according to the experiments.

5) Discussion of Decoder Design: In this section, we delve into the design of our network. Given that the primary objective of WSSS models is to extract additional information, such as affinity [46], from the provided labels, a simple network design is often preferred. In our implementation, we opt for a straightforward choice by employing a ViT-B as the encoder.

For the decoder part, we utilize a segmentation head comprising two dilated convolution layers with a 3×3 kernel size. Additionally, we explore two additional decoders focusing on multiscale cue extraction. The first is a pyramid decoder, which employs dilated convolutions at various scales, concatenating the results along the channel dimension. The second utilizes ASPP [53] for decoding. As depicted in Table VIII, our model achieves comparable performances across various computing resource metrics, such as parameters and GFLOPs. It is noteworthy that, despite the increased computational complexity resulting from the additional decoder designs, there is no observed improvement in performance. Consequently, we decide to retain the original design.

6) Effects of Different Label Supervision: To further explore the role of labels and the gap with their respective fully supervised methods, we insert segmentation heads to methods based on different backbones and conducted experiments on iSAID.

As depicted in Table IX, upon the provision of pixel-level labels, weakly supervised methods have all exhibited improvement, reaching a comparable level. However, approaches relying on WideResNet38 achieve only 53.42% and 60.75%, whereas ViT-based methods attain approximately 80% ratios. The disparities in performance ratios between fully supervised and weakly supervised methods suggest that the weakly supervised performance of ViT-based methods aligns more closely with their fully supervised counterparts, underscoring the potential of ViT in WSSS tasks.

V. CONCLUSION

This article introduces CTFA, a novel WSSS method that incorporates contrastive learning among tokens and label foreground activation. Recognizing the potential oversmoothing issue when directly applying ViT to WSSS tasks, especially in remote sensing images with large-area activation, we address this by leveraging tokens from two perspectives. In the patch-wise approach, we utilize relations extracted from the intermediate layer of ViT to supervise the output token representations. On the class-wise front, we ensure the consistency of global and local class-tokens to reveal entire object regions effectively. Additionally, we adopt a dual-branch decoder to simultaneously predict foreground and background, successfully activating the foreground region by fusing two kinds of probabilities.

By integrating these modules, we establish a single-stage WSSS framework. Extensive experiments conducted on the iSAID, Potsdam, and DeepGlobe datasets validate the effectiveness of our method, achieving new SOTA results in WSSS task.

REFERENCES

[1] Y. Du, Z. Fu, Q. Liu, and Y. Wang, "Weakly supervised semantic segmentation by pixel-to-prototype contrast," in *Proc. CVPR*, Jun. 2022, pp. 4320–4329.

[2] Z. Chen, T. Wang, X. Wu, X.-S. Hua, H. Zhang, and Q. Sun, "Class reactivation maps for weakly-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 969–978.

[3] J. Wang, Z. Zheng, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. NIPS*, 2021, pp. 1–16.

[4] A. Milioto, P. Lottes, and C. Stachniss, "Real-time semantic segmentation of crop and weed for precision agriculture robots leveraging background knowledge in CNNs," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 2229–2235.

[5] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[6] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[7] J. Ahn and S. Kwak, "Learning pixel-level semantic affinity with image-level supervision for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4981–4990.

[8] Y. Wang, J. Zhang, M. Kan, S. Shan, and X. Chen, "Self-supervised equivariant attention mechanism for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12275–12284.

[9] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.

[10] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 740–755.

[11] R. Zhou et al., "Weakly supervised semantic segmentation in aerial imagery via explicit pixel-level constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[12] Z. Li, X. Zhang, and P. Xiao, "One model is enough: Toward multiclass weakly supervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023.

[13] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2020, pp. 1–22.

[14] P.-T. Jiang, Y. Yang, Q. Hou, and Y. Wei, "L2G: A simple local-to-global knowledge transfer framework for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16886–16896.

[15] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.

[16] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023.

[17] P. Wang, W. Zheng, T. Chen, and Z. Wang, "Anti-oversmoothing in deep vision transformers via the Fourier domain analysis: From theory to practice," in *Proc. ICLR*, 2021, pp. 1–24.

[18] N. Park and S. Kim, "How do vision transformers work?" in *ICLR*, 2021, pp. 1–26.

[19] L. Ru, H. Zheng, Y. Zhan, and B. Du, "Token contrast for weakly-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3093–3102.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.

[20] C. Gong, D. Wang, M. Li, V. Chandra, and Q. Liu, "Vision transformers with patch diversification," 2021, *arXiv:2104.12753*.

[21] L. Xu, W. Ouyang, M. Bennamoun, F. Boussaid, and D. Xu, "Multi-class token transformer for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4310–4319.

[22] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9650–9660.

[23] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[24] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[25] J. Hu, Q. Wang, and X. Li, "Road extraction from satellite image via auxiliary road location prediction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2182–2185.

[26] J. Mei, R.-J. Li, W. Gao, and M.-M. Cheng, "CoANet: Connectivity attention network for road extraction from satellite imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 8540–8552, 2021.

[27] K. Liu, Z. Jiang, M. Xu, M. Perc, and X. Li, "Tilt correction toward building detection of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5854–5866, 2021.

[28] Z. Shao, P. Tang, Z. Wang, N. Saleem, S. Yam, and C. Sommai, "BRRNet: A fully convolutional neural network for automatic building extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 12, no. 6, p. 1050, Mar. 2020.

[29] Y. Liu, Z. Xiong, Y. Yuan, and Q. Wang, "Distilling knowledge from super-resolution for efficient remote sensing salient object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023.

[30] T.-S. Kuo, K.-S. Tseng, J.-W. Yan, Y.-C. Liu, and Y. F. Wang, "Deep aggregation net for land cover classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 252–256.

[31] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[32] Y. Liu, J. Yao, X. Lu, M. Xia, X. Wang, and Y. Liu, "RoadNet: Learning to comprehensively analyze road networks in complex urban scenes from high-resolution remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2043–2056, Apr. 2019.

[33] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[34] Z. Zhang, X. Huang, and J. Li, "DWin-HRFormer: A high-resolution transformer model with directional windows for semantic segmentation of urban construction land," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400714.

[35] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5617515.

[36] L. Ding, J. Zhang, and L. Bruzzone, "Semantic segmentation of large-size VHR remote sensing images using a two-stage multiscale training architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5367–5376, Aug. 2020.

[37] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[38] Z. Hu, B. Wang, and X. Li, "Densitytoken: Weakly-supervised crowd counting with density classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[39] B. Wang, Y. Zhao, and X. Li, "Multiple instance graph learning for weakly supervised remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[40] J. Li, B. Sun, S. Li, and X. Kang, "Semisupervised semantic segmentation of remote sensing images with consistency self-training," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[41] J. Chen, J. Zhu, Y. Guo, G. Sun, Y. Zhang, and M. Deng, "Unsupervised domain adaptation for semantic segmentation of high-resolution remote sensing imagery driven by category-certainty attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616915.

[42] Y. Jia, J. Gao, W. Huang, Y. Yuan, and Q. Wang, "Holistic mutual representation enhancement for few-shot remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023, Art. no. 5622613.

[43] Y. Wei and S. Ji, "Scribble-based weakly supervised deep learning for road surface extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[44] R. Lian and L. Huang, "Weakly supervised road segmentation in high-resolution remote sensing images using point annotations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[45] D. Zheng et al., "Utilizing bounding box annotations for weakly supervised building extraction from remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023.

[46] L. Ru, Y. Zhan, B. Yu, and B. Du, "Learning affinity from attention: End-to-end weakly-supervised semantic segmentation with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16846–16855.

[47] W. Zhang, S. Zeng, D. Wang, and X. Xue, "Weakly supervised semantic segmentation for social images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2718–2726.

[48] Z. Li, X. Zhang, P. Xiao, and Z. Zheng, "On the effectiveness of weakly supervised semantic segmentation for building extraction from high-resolution remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3266–3281, 2021.

[49] K. Fu et al., "WSF-NET: Weakly supervised feature-fusion network for binary segmentation in remote sensing image," *Remote Sens.*, vol. 10, no. 12, p. 1970, Dec. 2018.

[50] R. Zhou et al., "Weakly supervised semantic segmentation in aerial imagery via cross-image semantic mining," *Remote Sens.*, vol. 15, no. 4, p. 986, 2023.

[51] X. Zeng, T. Wang, Z. Dong, X. Zhang, and Y. Gu, "Superpixel consistency saliency map generation for weakly supervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023.

[52] J. Pan et al., "Learning self-supervised low-rank network for single-stage weakly and semi-supervised semantic segmentation," *Int. J. Comput. Vis.*, vol. 130, no. 5, pp. 1181–1195, May 2022.

[53] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[54] N. Araslanov and S. Roth, "Single-stage semantic segmentation from image labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4253–4262.

[55] L. Xu, M. Bennamoun, F. Boussaid, H. Laga, W. Ouyang, and D. Xu, "MCTformer+: Multi-class token transformer for weakly supervised semantic segmentation," 2023, *arXiv:2308.03005*.

[56] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[57] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[58] I. Potsdam. (2018). *2D Semantic Labeling Dataset*. [Online]. Available: http://www2.isprs.org

[59] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–181.

[60] P. Krähenbühl and V. Koltun, "Efficient inference in fully connected crfs with Gaussian edge potentials," in *Proc. NIPS*, vol. 24, 2011, pp. 1–9.

[61] J. Ahn, S. Cho, and S. Kwak, "Weakly supervised learning of instance segmentation with inter-pixel relations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2209–2218.

[62] J. Fan, Z. Zhang, T. Tan, C. Song, and J. Xiao, "CIAN: Cross-image affinity net for weakly supervised semantic segmentation," in *Proc. AAAI*, 2020, vol. 34, no. 7, pp. 10762–10769.

[63] Y.-T. Chang, Q. Wang, W.-C. Hung, R. Piramuthu, Y.-H. Tsai, and M.-H. Yang, "Weakly-supervised semantic segmentation via sub-category exploration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8991–9000.

[64] W. Liu, X. Kong, T.-Y. Hung, and G. Lin, "Cross-image region mining with region prototypical network for weakly supervised segmentation," *IEEE Trans. Multimedia*, vol. 25, pp. 1148–1160, 2023.

[65] D. Yu and S. Ji, "Long-range correlation supervision for land-cover classification from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4409814.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:44:41 UTC from IEEE Xplore. Restrictions apply.